

Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) states the resulting *precise reduction* of STEP-5B to two named gaps, and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection: the result is registered at T3 (proof sketch with named gaps).

2. Setting

Work at the corrected-convention production anchor ($\mu^2 = 0.005$, $r_{\text{braz}} = K(q_0) = \mu^2$; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) = \hat{r} + C(k^2 - q_0^2)^2, \quad \hat{r} = r_R + 2\lambda' I, \quad \lambda' = 3u + 30vM,$$

where $r_R = 0.3045257$, $M = M(\hat{r})$ self-consistently ($M_R = 0.109414$ at $I = 0$), $u = -0.86$, $v = +3.24$, $C = 1$, $q_0 = 0.6801748$, and $I = \sum_i A_i^2$ is the total condensate intensity of an admissible pattern $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$, $|q_i| = q_0$. Note $\lambda' = 8.0551 > 0$ at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part W of the Bloch Hessian in the plane-wave basis: entries w_t at momentum transfers $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$ with quartic-vertex weights

$$w_t = \lambda' \sum_{(i,j): \pm q_i \pm q_j = t} A_i A_j$$

(sextic $5vA^4$ -class transfers enter at $O(w_4)$; gap G2). Define $X = D^{-1/2} W D^{-1/2}$ and the weighted transfer norm $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$.

3. Two pattern-independent lemmas

Lemma 1 (Gershgorin–Schur row bound). *For the symmetric kernel X above, with $D_{\min} = \hat{r}$,*

$$\|X\| \leq \sup_k \sum_{t \neq 0} \frac{|w_t|}{\sqrt{D(k)D(k+t)}} \leq W_{\ell^1}(P) g(\hat{r}), \quad g(\hat{r}) = \frac{1}{\hat{r}} + \frac{1}{\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}}.$$

Proof. The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving $1/\hat{r}$; retaining additionally the degenerate-endpoint term $1/\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}$ of the Math434 LAM geometry makes g an upper envelope for every $|t| \leq 2q_0$ uniformly in the row, since any further leg sits at distance $\geq 64Cq_0^4$ above the gap in the quartic well. Summing over transfers yields the claim. $\square$

Lemma 2 (Second-order log-det envelope; isotropic trace identity). *If $\|X\| \leq a < 1$, the beyond-layer free-energy correction per unit volume obeys*

$$|\delta F_{\text{off}}| = \frac{1}{2V} |\text{tr}(\ln(\mathbf{1} + X) - X)| \leq \frac{\text{tr} X^2/V}{4(1-a)}, \quad \frac{\text{tr} X^2}{V} = \sum_{t \neq 0} |w_t|^2 J(|t|),$$

$$J(|t|) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{D(k)D(k+t)},$$

which depends on the pattern only through the moduli $|t|$ (isotropy of D).

Proof. Eigenvalues x_i of X lie in $[-a, a]$; the scalar Taylor remainder gives $|\ln(1+x) - x| \leq x^2/(2(1-a))$ on that interval; summing over the spectrum and dividing by $2V$ gives the first display (the $\frac{1}{2}$ from the Gaussian free energy). The trace identity is direct: $\text{tr} X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \rightarrow V \sum_t w_t^2 J(|t|)$, and J depends only on $|t|$ because D is a function of $|k|$ alone. The first off-diagonal order $\text{tr} X$ vanishes (no $t = 0$ entry in W). $\square$

4. The reduction (precise form of STEP-5B)

Proposition 1 (Pattern-generic reduction). *Let P be any admissible pattern with $a(P) := W_{\ell^1}(P)g(\hat{r}(I)) < 1$. Then*

$$|\delta F_{\text{off}}(P)| \leq \frac{W_{\ell^1}(P)^2 J_{\max}}{4(1-a(P))}, \quad J_{\max} := \max_{0 < |t| \leq 2q_0} J(|t|).$$

Consequently STEP-5B holds on the region $\{P : \text{RHS} < \text{interval margin}\}$, and full closure of the gate reduces to exactly two gaps:

- **G1 (weighted resonant ℓ^1 cap)**: a class-wide bound on $W_{\ell^1}(P)$ over the threat region. The Hartree self-penalty ($\hat{r} = r_R + 2\lambda I$) is self-limiting but insufficient alone ($a \rightarrow 6+$ as $I \rightarrow \infty$ at fixed mode count); the cap must combine (i) the layer's own large- I positivity (Math424 brackets confine the threat region to bounded I) and (ii) shell-resonance combinatorics limiting how much ℓ^1 weight can sit at small denominators.
- **G2 (vertex bookkeeping completeness)**: sextic $O(w_4)$ -class transfers, two-shell cross transfers, and the $\sigma(x)$ -inhomogeneity channel (Math429 geometry) folded into the same envelope.

5. Numerical verification

Script `codes/vacuum/beyond_layer_gershgorin_bound.py` (v1.0.1; 20/20 self-test asserts PASS; artefact `runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/result.json`).

Quantity	Value	Check
anchor $r_R, M_R, \lambda', 64Cq_0^4$ Math434 calibration ($A = 0.01$): $\hat{r}'$, row terms	0.3045257, 0.109414, 8.0551, 13.698 0.30613; $2.624 \times 10^{-3} + 3.879 \times 10^{-4}$	match locked constants reproduces $2.633 \times 10^{-3} + 3.893 \times 10^{-4}$; $\ X\ \leq 3.1 \times 10^{-3}$
$J(t)$, $ t /q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$	0.273, 0.218, 0.169, 0.133, 0.104	grid-refinement drift $< 6 \times 10^{-6}$; shell estimate 0.161 brackets $J_{\max}$
calibrated box $n_{\text{res}}=12$, $I = 4 \times 10^{-4}$	$a = 0.143$, $\delta \leq 2.38 \times 10^{-4}$	margin ratio $18.2 \times$
calibrated box $n_{\text{res}}=12$, $I = 10^{-3}$	$a = 0.347$, $\delta \leq 1.95 \times 10^{-3}$	margin ratio $2.2 \times$
LAM second order	$\delta_{\text{LAM}} = 6.7 \times 10^{-8}$	margin/64,000

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to $\leq 4 \times 10^{-6}$; J obeys the analytic shell estimate $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$ within a factor 1.7; dimensions: $[w] = [\hat{r}]$, $[J] = [\hat{r}]^{-2} [k]^3$ consistent with δF per unit volume.

The CALIBRATED boxes are stated regions, not derived caps — deriving the cap is exactly G1.

6. Devil’s-advocate (three objections)

- (α) “ W_{ℓ^1} grows without bound in the mode number n , so the reduction proves nothing class-wide.”
VALID-with-mitigation: this is precisely gap G1, registered rather than hidden. Mitigations already in hand: the threat region has bounded I (layer positivity brackets), the Hartree term raises $\hat{r}$ with I , and thin-spread patterns lose resonance gain in the layer functional itself (competition-closure generic-phase principle); none of these is yet assembled into a theorem, hence T3.
- (β) “ $J(|t|)$ is numerical only.” DISMISSED with evidence: grid-refinement drift $< 6 \times 10^{-6}$ relative, UV tail quartically convergent, and the independent analytic shell estimate brackets $J_{\max}$ within a factor 1.7.
- (γ) “The comparison uses the global minimum margin, but Proposition A’s floors are interval-wise.”
VALID-with-mitigation: using the weakest interval margin $+0.00432$ globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)
Precise statement: Lemmas A/B (rigorous) reduce STEP-5B to the explicit bound $|dF_{\text{off}}| \leq W_{l1}^2 J_{\max}/(4(1-a))$ with gaps G1 (class-wide W_{l1} cap) and G2 (vertex bookkeeping) named; constants certified at the production anchor (20/20 asserts).
Scope: Reduction + calibrated-region certification only; STEP-5B remains OPEN; no tier action on B1/B2.
Dependencies: A1-KERNEL-CONV (convention); margins from

	Proposition A (B2-PROPA-HLAYER chain).
Evidence grade:	ANALYTIC (lemmas) + EXECUTED (constants)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 20/20 PASS; result.json under runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/
Falsification gate:	An admissible pattern inside a certified box whose exact off-diagonal correction exceeds the Lemma-B envelope, or any pattern undercutting the +0.00432 band margin at the anchor.
Tier before / after:	(new claim) -- / T3
No-overclaim statement:	This note does NOT close STEP-5B and does NOT promote Reading-H; class-wide validity awaits G1+G2.
Next required action:	G1 -- assemble the I-cap (layer positivity) with shell-resonance combinatorics into a W _{l1} theorem.